

INT202 Complexity of Algorithms
Tutorial 4

1. Give the Θ notation on the solution to each of the following recurrences. You need to justify your answers.

a. $T(n) = 4T\left(\frac{n}{2}\right) + n$

b. $T(n) = 2T(\sqrt{n}) + \log n$

2. Give a tight asymptotic upper bound (Big-O notation) on the solution to the following function. You need to justify your answers.

```
T(int n){  
    if(n<=2) return 1  
    else return (T(floor(sqrt(n)))+1)  
}
```

3. Let set $S = \{S_k\}$, $k=1, 2, 3, 4$, denote a set of objects with weights and benefits as given in the table below.

k	1	2	3	4
b_k	25	15	20	36
w_k	7	2	3	6

What is an optimal solution to the fractional Knapsack problem for S assuming that we have a sack that can hold objects with total weight 10? Write down the answers with a brief analysis.

4. There are 4 items, 1 knapsack, and the capacity of the knapsack is 5. The values V of the 4 items are: 4 5 2 8 respectively, and the weights W of the four items are: 2 2 1 3 respectively. Solve the $\{0-1\}$ Knapsack problem with dynamic programming. You need to show the completed table with all computation results and procedures.